A **class** in Java is a **blueprint** or **template** from which **objects** are created.
Each object has:
Data (attributes) — what it *has*
Methods (functions) — what it *does*

- Car → Class (blueprint)
- c1 → Object
- color → Attribute
- start() → Method

```
class Car {
    String color;
    void start() {
        System.out.println("Car is starting...");
    }
}

public class Main {
    public static void main(String[] args) {
        Car c1 = new Car();   // object created
        c1.color = "Red";
        c1.start();
    }
}
```

Why do we use Classes?

1.Reusability – You can reuse the same class to create multiple objects. Example: You can create many cars (Car car1 = new Car(); Car car2 = new Car();).

2.Organization – Classes help organize your code better by grouping related variables and methods together.

3.Encapsulation – You can hide internal details and expose only necessary information (data protection).

4.Object-Oriented Programming (OOP) – Java is based on OOP principles (class, object, inheritance, polymorphism, etc.), and **class** is the foundation.

The General Form of a Class

```
class classname {

type instance-variable1;
type instance-variable2;
// ...
type instance-variableN;

type methodname1(parameter-list) {
// body of method
}


// ...
type methodnameN(parameter-list) {
// body of method
}
}
```

- Class defines a new data type. Once defined, this new type can be used to create objects of that type.
- A class is declared by use of the **class** keyword
- The data, or variables, defined within a **class** are called *instance variables.*
- The code is contained within *methods.*
- Collectively, the methods and variables defined within a class are called *members* of the class.
- Variables defined within a class are called instance variables

Class vs Object

Example — “House Blueprint”

A **class** is like an **architect’s blueprint** for a house.

It describes what the house *will have* — walls, doors, windows, rooms, etc.

An **object** is an **actual house** built from that blueprint.

Each house (object) looks similar (same structure) but can have **different colors, materials, or furniture** (different data values).

```
class House {
    String color;
    int rooms;

    void display() {
        System.out.println("Color: " + color + ", Rooms: " + rooms);
    }
}

public class Main {
    public static void main(String[] args) {
        House h1 = new House();   // Object 1
        h1.color = "Blue";
        h1.rooms = 3;

        House h2 = new House();   // Object 2
        h2.color = "Red";
        h2.rooms = 4;

        h1.display();
        h2.display();
    }
}
```

Concept	
Class	Blueprint or design plan
Object	Actual item built from the design
Attributes (variables)	Characteristics (e.g., color, size)
Methods (functions)	Actions the object can perform

A Simple Class

```
class Box {
double width;
double height;
double depth;
}
class BoxDemo {
public static void main(String args[]) {
Box mybox = new Box();
double vol;
// assign values to mybox's instance variables
mybox.width = 10;
mybox.height = 20;
mybox.depth = 15;
// compute volume of box
vol = mybox.width * mybox.height * mybox.depth;
System.out.println("Volume is " + vol);
}}
```

```
class Box {
double width;
double height;
double depth;
}
class BoxDemo2 {
public static void main(String args[]) {
Box mybox1 = new Box();
Box mybox2 = new Box();
double vol;
// assign values to mybox1's instance variables
mybox1.width = 10;
mybox1.height = 20;
mybox1.depth = 15;
/* assign different values to mybox2’s instance variables */
mybox2.width = 3;
mybox2.height = 6;
mybox2.depth = 9;
```

```
// compute volume of first box
vol = mybox1.width * mybox1.height * mybox1.depth;
System.out.println("Volume is " + vol);
// compute volume of second box
vol = mybox2.width * mybox2.height * mybox2.depth;
System.out.println("Volume is " + vol);
}
}
```

Declaring Objects

- When you create a class, you are creating a new data type.
- However, obtaining objects of a class is a two-step process.
- First, you must declare a variable of the class type. This variable does not define an object. Instead, it is simply a variable that can *refer* to an object.
- Second, you must acquire an actual, physical copy of the object and assign it to that variable. You can do this using the **new** operator.
- The **new** operator dynamically allocates (that is, allocates at run time) memory for an object and returns a reference to it.

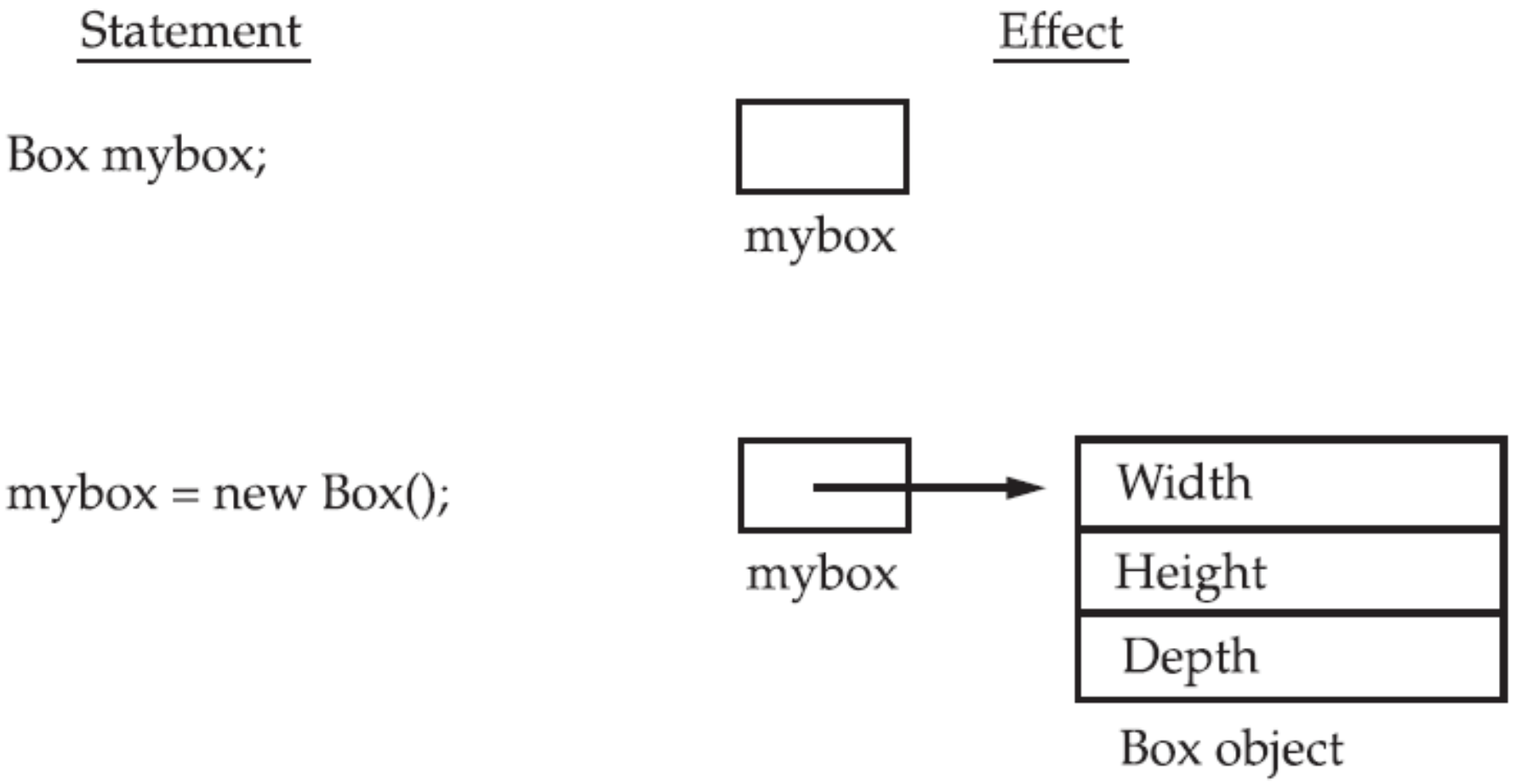


Figure 6-1 Declaring an object of type **Box**

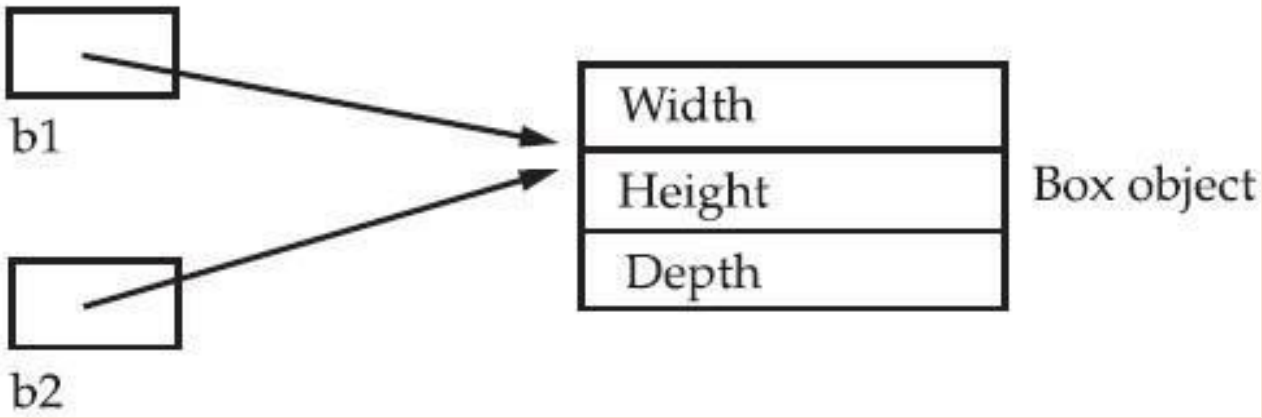
Box mybox = new Box();
This statement combines the two steps just described.
Box mybox; // declare reference to object
mybox = new Box(); // allocate a Box object

Assigning Object Reference Variables

Object reference variables act differently when an assignment takes place.

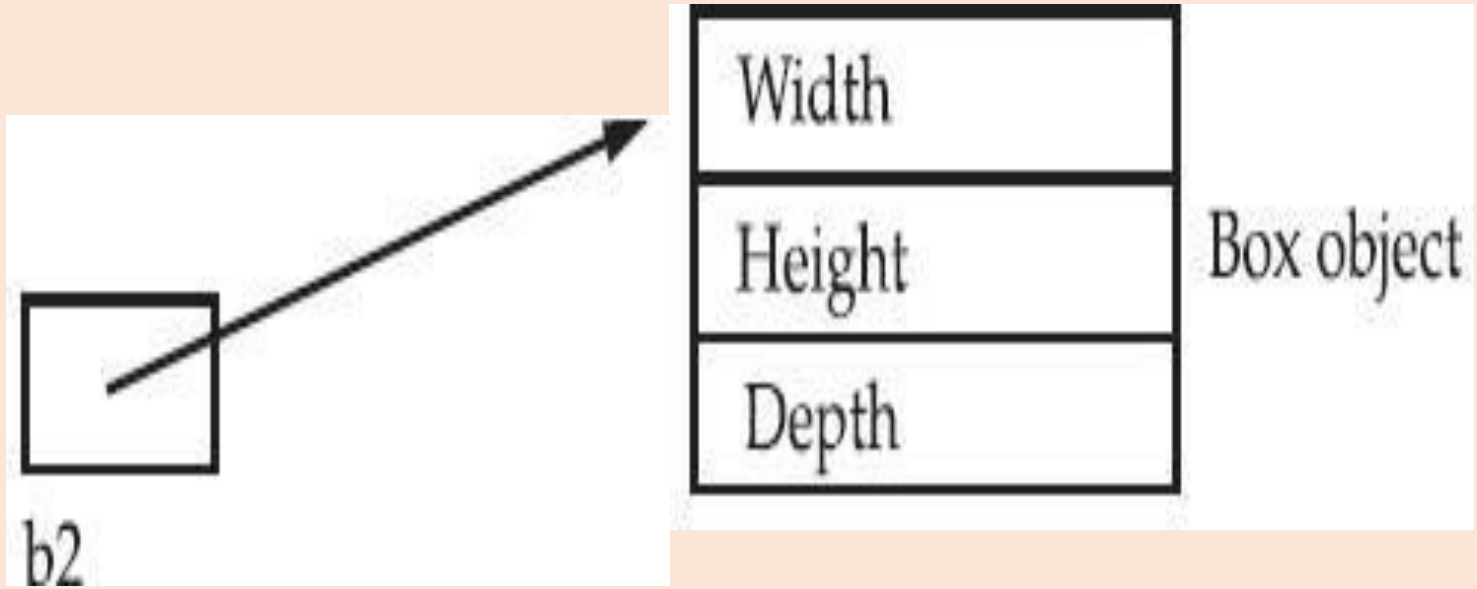
```
Box b1 = new Box();

Box b2 = b1;
```

This situation is depicted here:



```
Box b1 = new Box();
Box b2 = b1;
// ...
b1 = null;
```

Here, **b1** has been set to **null**, but **b2** still points to the original object



Introducing methods

This is the general form of a method:

type name(*parameter-list*) {
// body of method
}

Adding a Method to the Box Class

```
class Box {
    double width;
    double height;
    double depth;
    // display volume of a
    box void volume() {
        System.out.print("Volumeis ");
        System.out.println(width *
        height * depth);
    }
}
```

```
class BoxDemo3 {
    public static void main(String args[]) {

    Box mybox1 = new Box();

    // assign values to mybox1's instance variables

    mybox1.width = 10;
    mybox1.height = 20;
    mybox1.depth = 15;

    // display volume of first box

    mybox1.volume();
    }
}
```

Returning a Value

- While the implementation of **volume( )** does move the computation of a box's volume inside the **Box** class where it belongs, it is not the best way to do it.

```
class Box {
    double width;
    double height;
    double depth;
    // compute and return volume
     double volume() {
        return width * height * depth;
    }
}
```

```
class BoxDemo4 {
    public static void main(String args[]) {
    Box mybox1 = new Box();
     double vol;
        // assign values to mybox1's instance variables
        mybox1.width = 10;
        mybox1.height = 20;
        mybox1.depth = 15;
        // get volume of first box vol = mybox1.volume();
        System.out.println("Volume is " + vol);
    }
}
```

Adding a Method That Takes Parameters

- While some methods don't need parameters, most do. Parameters allow a method to be generalized.
- That is, a parameterized method can operate on a variety of data and/or be used in a number of slightly different situations

```
int square(int i)
{
    return i * i;
}

int x, y;
x = square(5); // x equals 25
x = square(9); // x equals 81
y = 2;
x = square(y); // x equals 4
```

```
class Box {
    double width;
    double height;
    double depth;
    // compute and return volume
    double volume() {
        return width * height * depth;
    }
    // sets dimensions of box
    void setDim(double w, double h, double d) {
        width = w;
        height = h;
        depth = d;
    }
}
```

```
class BoxDemo5 {

    public static void main(String args[]) {

    Box mybox1 = new Box();

    Box mybox2 = new Box();

    double vol;

        // initialize each box

        mybox1.setDim(10, 20, 15);
        mybox2.setDim(3, 6, 9);

        // get volume of first box

        vol = mybox1.volume();


        System.out.println("Volume is " + vol);

        // get volume of second box

        vol = mybox2.volume();
        System.out.println("Volume is " + vol);
    }}
```

Constructors

- It can be tedious to initialize all of the variables in a class each time an instance is created.
- Because the requirement for initialization is so common, Java allows objects to initialize themselves when they are created.
- This automatic initialization is performed through the use of a constructor.
- A *constructor* initializes an object immediately upon creation.
- It has the same name as the class in which it resides and is syntactically similar to a method.
- Once defined, the constructor is automatically called immediately after the object is created, before the **new** operator completes.

Constructors look a little strange because they have no return type, not even **void**

```
class Box {
    double width;
    double height;
    double depth;
    Box() {
        System.out.println("Constructing Box");
        width = 10;
        height = 10;
        depth = 10;
    }
    double volume() {
        return width * height * depth;
    }}
```

```
class BoxDemo6 {
    public static void main(String args[]) {
        // declare, allocate, and initialize Box objects
        Box mybox1 = new Box();
        Box mybox2 = new Box();
        double vol;
        // get volume of first box
        vol = mybox1.volume();
        System.out.println("Volume is " + vol);
        // get volume of second box
        vol = mybox2.volume();
        System.out.println("Volume is " + vol);
    }}
```

Parameterized Constructors

```
class Box {
    double width; double height; double depth;
    Box(double w, double h, double d) {
        width = w; height = h; depth = d;
    }
    double volume() {
        return width * height * depth;
    }
}
```

```
class BoxDemo7 {
    public static void main(String args[]) {
        // declare, allocate, and initialize Box objects
        Box mybox1 = new Box(10, 20, 15);
        Box mybox2 = new Box(3, 6, 9);
        double vol;
        vol = mybox1.volume();
        System.out.println("Volume is " + vol);
        vol = mybox2.volume();
        System.out.println("Volume is " + vol);
    }
}
```

The this keyword

- Sometimes a method will need to refer to the object that invoked it.
- To allow this, Java defines the **this** keyword. **this** can be used inside any method to refer to the *current* object

Box(double w, double h, double d) {

this.width = w; this.height = h; this.depth = d;

}

Uses of this:

→ To overcome shadowing or instance variable hiding.

→ To call an overload constructor

- can have local variables, including formal parameters to methods, which overlap with the names of the class' instance variables.
- However, when a local variable has the same name as an instance variable, the local variable *hides* the instance variable.

```
Box(double width, double height, double depth) {
    this.width = width; this.height = height;
    this.depth = depth;
}
```

Garbage Collection

- Since objects are dynamically allocated by using the **new** operator, you might be wondering how such objects are destroyed and their memory released for later reallocation.
- Java takes a different approach; it handles deallocation for you automatically.
- The technique that accomplishes this is called *garbage collection.*
- It works like this: when no references to an object exist, that object is assumed to be no longer needed, and the memory occupied by the object can be reclaimed.
- There is no explicit need to destroy objects as in C++.

The finalize( ) Method

- Sometimes an object will need to perform some action when it is destroyed. For example, if an object is holding some non-Java resource such as a file handle or character font, then you might want to make sure these resources are freed before an object is destroyed.
- To handle such situations, Java provides a mechanism called *finalization.*
- By using finalization, you can define specific actions that will occur when an object is just about to be reclaimed by the garbage collector.

The **finalize( )** method has this general form:

```
protected void finalize( )
{
// finalization code here
}
```

- the keyword **protected** is a specifier that prevents access to **finalize( )** by code defined outside its class.
- It is important to understand that **finalize( )** is only called just prior to garbage collection.

A Stack Class

```
class Stack {
    int stck[] = new int[10];
    int tos;
    // Initialize top-of-stack
    Stack() {
        top = -1;
    }
// Push an item onto the stack
void push(int item) {
    if(top==9)
        System.out.println("Stack is full.");
    else
        stck[++top] = item;
    }// Pop an item from the stack
```

```
    int pop() {
        if(top < 0) {
            System.out.println("Stack underflow.");
            return 0;
        }
        else
 return stck[top--];
}
}

class TestStack {
    public static void main(String args[]) {
    Stack mystack1 = new Stack();
    Stack mystack2 = new Stack();
```

```
        // push some numbers onto the stack
        for(int i=0; i<10; i++)
        mystack1.push(i);
        for(int i=10; i<20; i++)
        mystack2.push(i);
        // pop those numbers off the stack
        System.out.println("Stack in mystack1:");
         for(int i=0; i<10; i++)
              System.out.println(mystack1.pop());
              System.out.println("Stack in mystack2:");
              for(int i=0; i<10; i++)
              System.out.println(mystack2.pop());
}}
```

Overloading Methods

- In Java it is possible to define two or more methods within the same class that share the same name, as long as their parameter declarations are different.
- When this is the case, the methods are said to be *overloaded*, and the process is referred to as *method overloading*.
- Method overloading is one of the ways that Java supports polymorphism.
- Thus, overloaded methods must differ in the type and/or number of their parameters.
- While overloaded methods may have different return types, the return type alone is insufficient to distinguish two versions of a method.

```
class OverloadDemo {
    void test() {
        System.out.println("No parameters");
    }
    void test(int a) {
        System.out.println("a: " + a);
    }
    void test(int a, int b) {
        System.out.println("a and b: " + a + " " + b);
    }
    double test(double a) {
        System.out.println("double a: " + a);
        return a*a;
    }}
```

```
class Overload {
    public static void main(String args[]) {
     OverloadDemo ob = new OverloadDemo();
     double result;
        // call all versions of test()
        ob.test();
        ob.test(10);
        ob.test(10, 20);
        result = ob.test(123.25);
        System.out.println("Result of ob.test(123.25): " + result);
    }
}
```

```
class MathOperations {

    // Method 1: add two integers
    int add(int a, int b) {
        return a + b;
    }

    // Method 2: add three integers
    int add(int a, int b, int c) {
        return a + b + c;
    }

    // Method 3: add two double numbers
    double add(double a, double b) {
        return a + b;
    }
}
```

```
public class OverloadingExample {
    public static void main(String[] args) {
        MathOperations math = new MathOperations();


        System.out.println("Sum of two integers: " + math.add(10, 20));
        System.out.println("Sum of three integers: " + math.add(10, 20, 30));
        System.out.println("Sum of two doubles: " + math.add(5.5, 6.7));
    }
}
```

- When an overloaded method is called, Java looks for a match between the arguments used to call the method and the method’s parameters.

- However, this match need not always be exact. In some cases, Java’s automatic type conversions can play a role in overload resolution.

For example, consider the following program:

```
// Automatic type conversions apply to overloading. class

OverloadDemo {
    void test() {
        System.out.println("No parameters");
    }

    void test(int a, int b) {
        System.out.println("a and b: " + a + " " + b);
    }

    void test(double a) {
        System.out.println("Inside test(double) a: " + a);
    }
}
```

```
class Overload {
    public static void main(String args[]) {
        OverloadDemo ob = new OverloadDemo();
         int i = 88;
        ob.test();
        ob.test(10, 20);
        ob.test(i); // this will invoke test(double)
        ob.test(123.2); // this will invoke test(double)
    }
}
```

Overloading Constructors

- In addition to overloading normal methods, you can also overload constructor methods. In fact, for most real-world classes that you create, overloaded constructors will be the norm, not the exception.

```
class Box {
    double width;
    double height;
    double depth;
    // constructor used when all dimensions specified
    Box(double w, double h, double d) {
        width = w; height = h; depth = d;
    }
```

```
    // constructor used when no dimensions specified

    Box() {
        width = -1; // use -1 to indicate
        height = -1; // an uninitialized
        depth = -1; // box
    }

    // constructor used when cube is created

    Box(double len) {
        width = height = depth = len;
    }

    // compute and return volume

    double volume() {
        return width * height * depth;
    }
}
```

```
class OverloadCons {
    public static void main(String args[]) {
        // create boxes using the various constructors
        Box mybox1 = new Box(10, 20, 15);
        Box mybox2 = new Box();
        Box mycube = new Box(7);
        double vol;
        vol = mybox1.volume();
        System.out.println("Volume of mybox1 is " + vol);
        // get volume of second box
         vol = mybox2.volume();
        System.out.println("Volume of mybox2 is " + vol);
```

```
        // get volume of cube

        vol = mycube.volume();
        System.out.println("Volume of mycube is " + vol);
    }
}
```

Using Objects as Parameters

- So far, we have only been using simple types as parameters to methods. However, it is both correct and common to pass objects to methods.
- For example, consider the following short program:

```
// Objects may be passed to methods. class Test {
    int a, b;
    Test(int i, int j) {
        a = i;
        b = j;
    }
    // return true if o is equal to the invoking object
    boolean equals(Test o) {
        if(o.a == a && o.b == b)
        return true;
        else
         return false;
}}
```

```
class PassOb {
    public static void main(String args[]) {
    Test ob1 = new Test(100, 22);
    Test ob2 = new Test(100, 22);
    Test ob3 = new Test(-1, -1);
        System.out.println("ob1 == ob2: " + ob1.equals(ob2));
        System.out.println("ob1 == ob3: " + ob1.equals(ob3));
    }
}
```

This program generates the following output:

```
 ob1 == ob2: true
ob1 == ob3: false
```

Argument Passing

- In general, there are two ways that a computer language can pass an argument to a subroutine.
- The first way is ***call-by-value.*** This approach copies the *value* of an argument into the formal parameter of the subroutine. Therefore, changes made to the parameter of the subroutine have no effect on the argument.
- The second way an argument can be passed is ***call-by-reference***. In this approach, a reference to an argument (not the value of the argument) is passed to the parameter.
- Inside the subroutine, this reference is used to access the actual argument specified in the call. This means that changes made to the parameter will affect the argument used to call the subroutine

```
class Test {
        void meth(int i, int j) {
                i *= 2;
                j /= 2;

        }
}
class CallByValue {
        public static void main(String args[]) {
        Test ob = new Test();
                int a = 15, b = 20;
                System.out.println("a and b before call: " + a + " " + b);
                ob.meth(a, b);
                System.out.println("a and b after call: " + a + " " + b);
        }
}
```

output = a and b before call: 15 20 a and b after call: 15 20

- When you pass an object to a method, the situation changes dramatically, because objects are passed by what is effectively call-by-reference.
- Thus, when you pass this reference to a method, the parameter that receives it will refer to the same object as that referred to by the argument.
- This effectively means that objects are passed to methods by use of call-by-reference. Changes to the object inside the method *do* affect the object used as an argument.

```
class Test {
        int a, b;
        Test(int i, int j) {
                a = i;
                b = j;
}// pass an object
void meth(Test o) {
o.a*=2;
o.b/=2;
}}
class CallByRef {
        public static void main(String args[]) {
         Test ob = new Test(15, 20);
                System.out.println("ob.a and ob.b before call: " + ob.a + " " + ob.b);
                ob.meth(ob);
                System.out.println("ob.a and ob.b after call: " + ob.a + " " + ob.b);
        }
}
```

This program generates the following output:

ob.a and ob.b before call: 15 20

ob.a and ob.b after call: 30 10

Returning Objects

- A method can return any type of data, including class types that you create. For example, in the following program, the **incrByTen( )** method returns an object in which the value of **a** is ten greater than it is in the invoking object.

```
class Test {
    int a;
    Test(int i) {
        a = i;
    }
    Test incrByTen() {
        Test temp = new Test(a+10);

        return temp;
    }
}
```

```
class RetOb {
    public static void main(String args[]) {
    Test ob1 = new Test(2);
        Test ob2;
        ob2 = ob1.incrByTen();
        System.out.println("ob1.a: " + ob1.a);
        System.out.println("ob2.a: " + ob2.a);
        ob2 = ob2.incrByTen();
        System.out.println("ob2.a after second increase: " + ob2.a);
    }
}
```

The output generated by this program is shown here:

```
ob1.a: 2
ob2.a: 12
ob2.a after second increase: 22
```

Recursion

- Java supports *recursion*. Recursion is the process of defining something in terms of itself. A method that calls itself is said to be *recursive*.
- The classic example of recursion is the computation of the factorial of a number. The factorial of a number *N* is the product of all the whole numbers between 1 and *N*.
- For example, 3 factorial is 1 · 2 · 3, or 6. Here is how a factorial can be computed by use of a recursive method:

```
class Factorial {
        int fact(int n) {
            int result;
            if(n==1)
             return 1;
            result = fact(n-1) * n;
            return result;
        }
}
class Recursion {
        public static void main(String args[]) {
         Factorial f = new Factorial();
            System.out.println("Factorial of 3 is " + f.fact(3));
            System.out.println("Factorial of 4 is " + f.fact(4));
            System.out.println("Factorial of 5 is " + f.fact(5));
            }}
```

Introducing Access Control

- Encapsulation provides another important attribute: *access control.*
- Through encapsulation, you can control what parts of a program can access the members of a class.
- By controlling access, you can prevent misuse. For example, allowing access to data only through a well defined set of methods, you can prevent the misuse of that data.
- Java's access specifiers are **public**, **private**, and **protected**.
- **protected** applies only when inheritance is involved
- When a member of a class is modified by the **public** specifier, then that member can be accessed by any other code.
- When a member of a class is specified as **private**, then that member can only be accessed by other members of its class.
- Now you can understand why **main( )** has always been preceded by the **public** specifier. It is called by code that is outside the program—that is, by the Java run-time system.
- When no access specifier is used, then by default the member of a class is public within its own package, but cannot be accessed outside of its package.
- An access specifier precedes the rest of a member's type specification. That is, it must begin a member's declaration statement. Ex public int I; private int j;

```
class Test {
                int a; // default access public

                    int b; // public access
    private int c; // private access

    void setc(int i) {

    c = i;
    }

    int getc() { // get c's value

     return c;


    }
}
```

```
class AccessTest {

    public static void main(String args[]) {

    Test ob = new Test();

        ob.a = 10;
        ob.b = 20;
        // This is not OK and will cause an error
        // ob.c = 100; // Error!

        // You must access c through its methods

         ob.setc(100); // OK
        System.out.println("a, b, and c: " + ob.a + " " + ob.b + " " +
        ob.getc());
    }
}
```

- Outside of the class in which they are defined, **static** methods and variables can be used independently of any object.
- To do so, you need only specify the name of their class followed by the dot operator.
- For example, if you wish to call a **static** method from outside its class, you can do so using the following general form:

  ***classname.method*( )**

- Here, *classname* is the name of the class in which the **static** method is declared. As you can see, this format is similar to that used to call non-**static** methods through object-reference variables.
- A **static** variable can be accessed in the same way—by use of the dot operator on the name of the class. This is how Java implements a controlled version of global methods and global variables.
- Here is an example. Inside **main( )**, the **static** method **callme( )** and the **static** variable **b** are accessed through their class name **StaticDemo**.

```
class StaticDemo {
static int a = 42;
static int b = 99;
 static void callme() {

System.out.println("a = " + a);
}
}
class StaticByName {
public static void main(String args[]) {
StaticDemo.callme();
System.out.println("b = " + StaticDemo.b);
}
}
```

Here is the output of this program:

```
a = 42
b = 99
```

Introducing final

- A variable can be declared as **final**. Doing so prevents its contents from being modified. This means that you must initialize a **final** variable when it is declared.
- For example:

```
final int FILE_NEW = 1;

final int FILE_OPEN = 2;

final int FILE_SAVE = 3;

final int FILE_SAVEAS = 4;

final int FILE_QUIT = 5;
```

- Subsequent parts of your program can now use **FILE_OPEN**, etc., as if they were constants, without fear that a value has been changed.
- It is a common coding convention to choose all uppercase identifiers for **final** variables.
- Variables declared as **final** do not occupy memory on a per-instance basis. Thus, a **final** variable is essentially a constant.
- The keyword **final** can also be applied to methods, but its meaning is substantially different than when it is applied to variables.

Introducing Nested and Inner Classes

- It is possible to define a class within another class; such classes are known as *nested classes*.
- The scope of a nested class is bounded by the scope of its enclosing class.
- Thus, if class B is defined within class A, then B does not exist independently of A.
- A nested class has access to the members, including private members, of the class in which it is nested. However, the enclosing class does not have access to the members of the nested class.
- There are two types of nested classes: static and non-static.
- A static nested class is one that has the static modifier applied. Because it is static, it must access the non-static members of its enclosing class through an object

Static Nested Class

This is a **static class inside another class**.

✔ **Properties:**

- Does **not** need an object of outer class
- Can access **only static members** of outer class
- Works like a normal class but inside another class

```
class Outer {
    static class Inner {
        void show() {
            System.out.println("Inside Static Nested Class");
        }
    }
}

public class Test {
    public static void main(String[] args) {
        Outer.Inner obj = new Outer.Inner();
        obj.show();
    }
}
```